

Eigenvalues and EigenVectors

Eigenvectors are vectors that keep their direction.

When a matrix transforms a vector, the vector usually changes both **direction and magnitude**. But some special vectors only get **stretched or compressed**. These are **eigenvectors**.

For a square matrix **A**, an Eigenvector and Eigenvalue make this equation true:

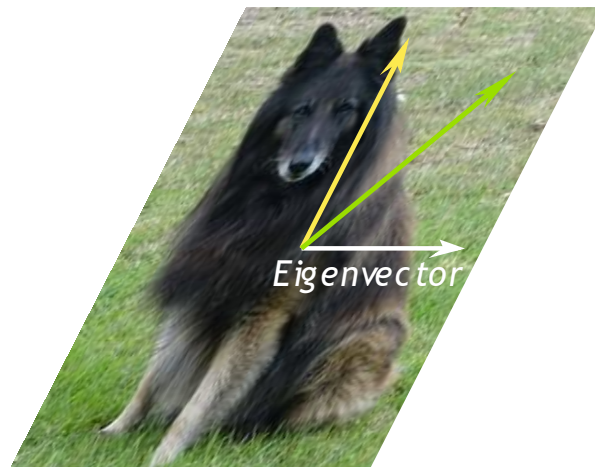
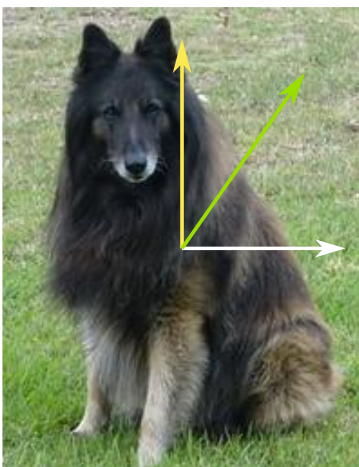
$$Av = \lambda v$$

Where:

A = matrix , v = **eigenvector** , λ = **eigenvalue**.

Key idea:

The matrix transforms the vector, but its **direction stays unchanged**.



Eigenvectors : An Example

For this matrix :

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}$$

an **eigenvector** is

$$\begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

with a matching **eigenvalue** of 6.

This satisfies the equation - **$\mathbf{Av} = \lambda\mathbf{v}$**

$\mathbf{Av}$ gives us:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

$\lambda\mathbf{v}$ gives us :

$$6 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

LHS = RHS.

But How do we find these Eigenvalues and Vectors? 🤔

Finding Eigenvalues

First, put in an Identity Matrix in the discussed equation:

$$\mathbf{Av} = \lambda \mathbf{Iv}$$

$\mathbf{I}$ is an Identity matrix (has 1s on the main diagonal and 0s everywhere else).

Bring all to left hand side:

$$\mathbf{Av} - \lambda \mathbf{Iv} = 0$$

Solve for λ using just the determinant

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

Using previous example :

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -6 - \lambda & 3 \\ 4 & 5 - \lambda \end{bmatrix} = 0$$

Calculating that determinant gets:

$$(-6 - \lambda)(5 - \lambda) - 3 \times 4 = 0$$

$$\Rightarrow \lambda^2 + \lambda - 42 = 0$$

$$\Rightarrow \lambda = -7 \text{ or } 6$$

Find the Eigenvector

Start with:

$$\mathbf{Av} = \lambda \mathbf{v}$$

Put in the values from before:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 6 \begin{bmatrix} x \\ y \end{bmatrix}$$

After multiplying we get these two equations:

$$-6x + 3y = 6x$$

$$4x + 5y = 6y$$

Bringing all to left hand side:

$$-12x + 3y = 0$$

$$4x - y = 0$$

Either equation reveals:

$$y = 4x$$

So the **eigenvector** is any non-zero multiple of this:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Why Do AI & ML Care?

Eigenvalues and eigenvectors help identify the **important directions in data**.

PCA

Principal Component Analysis uses eigenvectors of the **covariance matrix** to find directions of maximum variance.

Eigenvectors → principal directions

Eigenvalues → amount of variance captured

This helps with:

Dimensionality Reduction → Noise Reduction → Visualization → Feature Analysis

Eigenvalue methods also appear in:

- Spectral clustering
- Graph-based ML
- Computer vision
- Recommendation systems
- Optimization.

Resources To Learn Further

LibreTexts

Clear textbook-style explanations with worked examples.

Math Is Fun

Beginner-friendly visual explanations of matrices and related concepts.

3Blue1Brown - Essence of Linear Algebra

Excellent visual intuition for eigenvectors, eigenvalues, and transformations.

Khan Academy - Eigenvalues & Eigenvectors

Step-by-step explanations and practice problems.

MIT OpenCourseWare - Linear Algebra

More rigorous treatment for building a strong mathematical foundation.